



Chapter S1A: Permutations and Combinations

SYLLABUS INCLUDES

- addition and multiplication principles for counting;
- concepts of permutation $({}^nP_r)$ and combination $({}^nC_r)$;
- arrangements of objects in a line or in a circle, including cases involving repetition and restriction.

PRE-REQUISITES

- Basic Set Theory

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1 INTRODUCTION

Is an alphanumeric password with at least one capital letter really more secure than one without? Is it possible for you to eat something different for lunch every day in your two years in RI? If you have at any time pondered about these questions, you will find the necessary tools to help you arrive at the answers in this chapter. The study of permutations and combinations is essentially an exercise in systematic counting, and is a fundamental skill for those seeking to study higher level *Combinatorial Analysis*. It has a wide range of applications – from programming, sampling and the allocation of resources to the study of number theory.

2 TWO COUNTING PRINCIPLES

There are two basic principles of counting which are instrumental to our investigations in this chapter.

2.1 The Addition Principle

Let X and Y be two actions that cannot occur simultaneously. We say that X and Y are mutually exclusive. If X can occur in x different ways, and Y can occur in y different ways, then X **or** Y can occur in $(x + y)$ ways.

Note: *The Addition Principle can be extended to any finite number of actions that are mutually exclusive.*

Example 1

In order to travel from Outram to Yishun, there are 2 possible routes by MRT and 3 possible routes by bus. How many routes are there from Outram to Yishun?

Solution

Case 1: Travel by MRT from Outram to Yishun.

Number of ways = 2.

Case 2: Travel by Bus from Outram to Yishun.

Number of ways = 3.

Total number of routes from Outram to Yishun = $2 + 3 = 5$.

2.2 The Multiplication Principle

Let P and Q be two actions. If P can occur in p different ways and Q can occur in q different ways, then P **and** Q can occur in succession in $(p \times q)$ ways.

Note: The Multiplication Principle can be extended to any finite number of actions occurring in succession.

Example 2

To get to school from home, John has to first travel to an MRT station either by walking, or taking one of the two available feeder bus services. After that, he could either travel on the MRT by taking either the North South Line from Dhoby Ghaut to Bishan, or the Circle Line from Dhoby Ghaut to Bishan. How many routes are there from home to school?

Solution

Stage 1: Travel from home to MRT station.

Number of ways = 3.

Stage 2: Travel from MRT station to school.

Number of ways = 2.

Number of routes from home to school = $3 \times 2 = 6$.

3 PERMUTATIONS

A **permutation** is an arrangement of objects in which the order is important.

For instance, the numbers 123, 132, 213, 231, 312, 321 are the different permutations of the digits 1, 2, 3 and the numbers 113, 131, 311 are the different permutations of the digits 1, 1 and 3.

3.1 Permutations of Distinct Objects

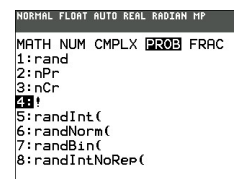
Result 1

The number of permutations of n distinct objects, taken ALL at a time without replacement, is given by $n! = n(n-1)(n-2)\dots(3)(2)(1)$, for $n \in \mathbb{Z}^+$.

This is the **factorial** of n .
Note: $0! = 1$.

GC Commands

The commands needed for this chapter can be found in the **PROB** section of the **MATH** menu, as shown in the figure to the right



OR

using the shortcut key $\boxed{\alpha}\boxed{\text{window}}\boxed{\text{f2}}$

**Example 3**

- (i) How many ways can CAT be arranged to obtain different codewords?
 (ii) How many ways can CATS be arranged to obtain different codewords?

Solution

(i) Number of ways = $3! = 6$.

By systematic listing:

CAT ACT TAC
 CTA ATC TCA

(ii) Number of ways = $4! = 24$.

An easy way to understand Result 1 is to consider objects being placed into slots. In Example 3 above, we want to permute the letters of the word “CAT” to get different codewords. Imagine we have 3 slots to place our 3 distinct letters. For the first slot, there are 3 letters to choose from. After we have made our choice, we now have only 2 letters left to choose from for the second slot. Finally, for the last slot, we have only 1 letter left to choose from. Since the actions of placing the letters into the slots occur one after another, using the Multiplication Principle, the number of permutations = $3 \times 2 \times 1 = 3! = 6$.

3.2 Permutations of Objects taken from Sets of Distinct Objects**Result 2**

The number of permutations of n DISTINCT objects, taken r at a time without replacement, is given by ${}^nP_r = \frac{n!}{(n-r)!}$, where $0 \leq r \leq n$.

Question:

What is nP_n ?

Using the same idea of objects being placed into slots, it is not difficult to see how we arrive at the above result. Consider the permutations of 4 letters of the word “NUMBER”. For the first slot, we have all 6 letters to choose from. After making our choice, we now have 5 letters left for the second slot. Subsequently, we have 4 and 3 letters left to choose from for the third and fourth slots respectively.

Applying the Multiplication Principle, the number of permutations = $6 \times 5 \times 4 \times 3 = 360$.

Note: $6 \times 5 \times 4 \times 3 = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = \frac{6!}{2!} = \frac{6!}{(6-4)!}$, which is denoted by 6P_4 .



Example 4

A man has 5 different books and wishes to give one to each of his 3 daughters. In how many ways can he do this?

Solution

Number of ways = ${}^5P_3 = 60$ (or $5 \times 4 \times 3 = 60$).

3.3 Permutations of Objects taken with Replacement**Result 3**

The number of permutations of n DISTINCT objects, taken r at a time with replacement, is given by n^r .

Consider the same problem of permutations of 4 letters of the word “NUMBER”, but this time, we allow the letters to be repeated. For the first slot, we have all 6 letters to choose from. Since we allow repetition, the second slot can again be any of the 6 letters. Similarly, we still have 6 choices each for the third and fourth slots.

Applying the Multiplication Principle, the number of permutations = $6 \times 6 \times 6 \times 6 = 6^4$
 $= 1296$.

Example 5

How many 3-digit numbers can be formed from the set $\{1, 2, 3, 4, 5\}$ if the digits may be repeated?

Solution

Since the digits may be repeated, we have 5 choices for every digit of the 3-digit number.

Number of 3-digit numbers that can be formed = $5 \times 5 \times 5 = 5^3 = 125$.

3.4 Permutations Involving Non-Distinct Objects**Result 4**

The number of permutations of n objects, taken all at a time without replacement, of which n_1 are of the 1st type, n_2 of the 2nd type, ..., n_k of the k^{th} type, where $n_1 + n_2 + \dots + n_k = n$ is given by $\frac{n!}{n_1!n_2!\dots n_k!}$.

**Refer to Appendix A for an explanation of the above result*

Example 6

Find the number of arrangements of all the letters of the word “MISSISSIPPI” in which

- (i) there is no restrictions,
- (ii) the four letters I are always together,
- (iii) M and two letters P are always together.

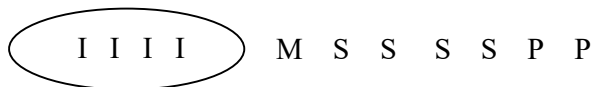
Solution

There are 1 M, 4 I's, 4 S's and 2 P's and altogether 11 letters.

(i) Number of ways = $\frac{11!}{1!4!4!2!} = 34\,650$.

(ii) Grouping Method

Since the 4 I's are always together, we consider them as 1 unit.
Therefore, any arrangement would be like the following.

 I I I I M S S S S P P

Take note of this
“grouping together
as 1 unit” method.

Number of ways to arrange this unit and the other 7 letters in a row = $\frac{8!}{4!2!} = 840$.

(iii) Grouping Method

Since M and 2 P's are always together, we consider them as 1 unit.
Therefore, any arrangement would be like the following.

 Unit of M and 2 P's I I I I S S S S

Number of ways to arrange this unit and the other 8 letters in a row = $\frac{9!}{4!4!}$.

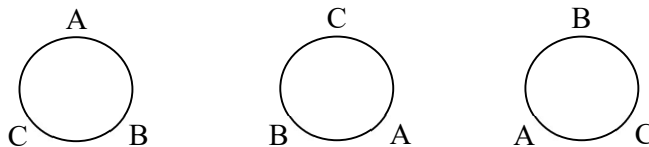
The number of ways to arrange the M and 2 P's within the unit itself = $\frac{3!}{2!}$

So required number of ways = $\frac{9!}{4!4!} \times \frac{3!}{2!} = 1890$.

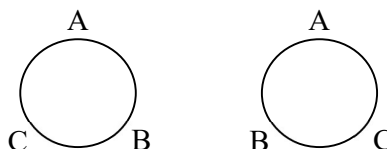
3.5 Circular Permutations

A circular permutation is an arrangement of objects in a circle. Two circular arrangements are considered to be identical if one can be rotated to form the other.

Hence the following circular permutations of 3 objects are identical,



but the following circular permutations are **not** identical.



Result 5

The number of permutations of n DISTINCT objects in a circle is given by $\frac{n!}{n} = (n-1)!$.

Refer to **Appendix B for an explanation of the above result.*

Example 7

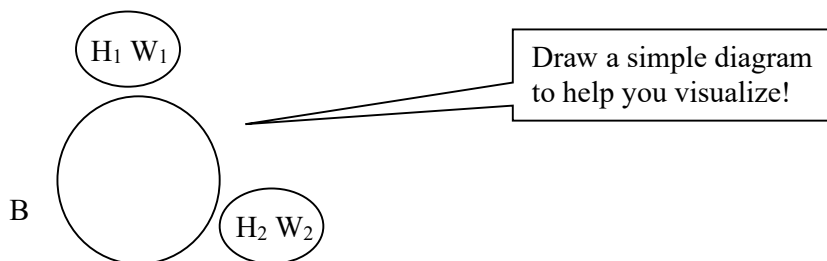
- (a) How many ways can 2 couples and a boy be seated at a round table?
 (b) How many ways can 2 couples and a boy be seated at a round table if each couple is to be seated together?

Solution

(a) Number of ways = $\frac{5!}{5} = (5-1)! = 24$.

(b) Grouping method

Since each couple is to be seated together, we consider them as 1 unit



Number of ways to arrange the 2 units and the boy at the round table = $\frac{3!}{3}$

Number of ways to arrange the first couple within the unit = $2!$

Number of ways to arrange the second couple within the unit = $2!$

$\therefore$ Required number of ways = $\frac{3!}{3} \times 2! \times 2! = 8$

Note: When the seats are numbered, the number of arrangements will be different. Why?

In this case, two circular arrangements such that one can be obtained by rotating the other will no longer be identical as the seats are now numbered.

Recalculating Example 7, considering the seats are numbered,

(a) number of ways = $24 \times 5 = 120$

(b) number of ways = $8 \times 5 = 40$

In a circular permutation,

No of ways for **numbered seats**

= No of ways for **unnumbered seats** $\times$ No of seats

4 COMBINATIONS

A **combination** is a selection of objects in which the order of selection does not matter.

Result 6

The number of combinations of n DISTINCT objects, taken r at a time without replacement, is given by nC_r or $\binom{n}{r}$, where $0 \leq r \leq n$.

Note: The symbols nC_r and $\binom{n}{r}$ are read “ n choose r ”.

4.1 Relationship between nP_r and nC_r

Let us revisit the problem we encountered in Example 4:

A man has 5 different books and wishes to give one to each of his 3 daughters. In how many ways can he do this?

In Example 4, we have already found that the number of ways is ${}^5P_3 = 60$ ways.

Alternatively, let us solve the problem by considering a sequence of events:

First, we select 3 books out of the 5 available books.

Next, we distribute the 3 chosen books to the daughters such that each daughter gets one book.

By result 6, there are 5C_3 ways to select 3 books out of 5 available books. There are $3!$ ways to distribute the chosen books to the daughters.

By multiplication principle, there are ${}^5C_3 \times 3! = 60$ ways.

Thus, we can see that ${}^5P_3 = {}^5C_3 \times 3!$.

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5C_3

Generalising, we have the following:

$$(i) \quad {}^nP_r = {}^nC_r \times r!,$$

$$(ii) \quad {}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!}, \quad \text{where } 0 \leq r \leq n.$$

Note : $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ is given in the Formula List.

Note:

$$1) \quad \binom{n}{0} = 1 \text{ and } \binom{n}{n} = 1. \text{ By convention, we define } \binom{0}{0} = 1.$$

$$2) \quad \binom{n}{r} = \binom{n}{n-r} \text{ where } 0 \leq r \leq n.$$

Example 8

Find the number of ways in which 6 people can be divided into

- (i) 2 groups consisting of 4 and 2 people;
- (ii) 2 groups of 3 people each.

Solution

- (i) Since there are 6 people in total, if we were to choose 4 people to form a group, the remaining 2 will automatically form the 2nd group.

Hence, we need only find the number of ways to choose a group of 4 people.

$$\begin{aligned}
 &\text{Number of ways in which 6 people can be divided into 2 groups consisting of} \\
 &\text{4 and 2 people} \\
 &= \text{Number of ways to choose a group of 4 people} \\
 &= {}^6C_4 \\
 &= 15
 \end{aligned}$$

We can also choose a group of 2 people first. Why?

- (ii) Number of ways to choose a group of 3 people = 6C_3 .

The remaining 3 people will automatically form the 2nd group.

However, 2 groups with the same size contribute to double-counting.

Number of ways in which 6 people can be divided into 2 groups of 3 people

$$\text{each} = \frac{{}^6C_3}{2!} = 10$$

Why divide by 2! ?

Extension:

Find the number of ways in which 6 people can be divided into 3 groups of 2 people each.

$$\text{Number of ways} = \frac{{}^6C_2 {}^4C_2}{3!} = 15$$

5 FURTHER TECHNIQUES AND PROBLEMS

This section will illustrate some common techniques used to solve problems involving permutations and combinations. Note that the more complicated problems are best tackled using the Addition and Multiplication Principles in conjunction with systematic analysis of cases.

These techniques include:

- grouping method
- complement method
- slotting method
- systematic listing

Example 9

How many integers between 3000 and 5000 can be formed using only the digits 1, 2, 3, 4, 5 if

- no digit may be repeated?
- the digits can be repeated?
- no repetition of the digits is allowed and the integers formed must be even?

Solution

- (i) Number of choices for 1st digit = 2.
Number of choices for 2nd digit = 4.
Number of choices for 3rd digit = 3.
Number of choices for 4th digit = 2.

Total number of integers that can be formed = $2 \times 4 \times 3 \times 2 = 48$.

- (ii) Number of choices for 1st digit = 2.
Number of choices for 2nd, 3rd, 4th digits = 5.

Total number of integers that can be formed = $2 \times 5 \times 5 \times 5 = 250$.

'3' or '4'

Note the use of
Multiplication
Principle here.

(iii) Since integers formed must be even, the last digit can only be 2 or 4.

Case 1: The last digit is '2'.

Number of choices for 1st digit = 2.

Number of choices for 2nd digit = 3.

Number of choices for 3rd digit = 2.

Number of integers that can be formed = $2 \times 3 \times 2 = 12$.

'3' or '4'

Case 2: The last digit is '4'.

Number of choices for 1st digit = 1.

Number of choices for 2nd digit = 3.

Number of choices for 3rd digit = 2.

Number of integers that can be formed = $1 \times 3 \times 2 = 6$.

Can only be '3'

Total number of integers that can be formed = $12 + 6 = 18$.

Note the use of
the Addition
Principle here.

Can you think of an alternative method to solve this?

Example 10

How many ways can 3 girls and 2 boys be arranged in a row so that

- (i) the boys are always together?
- (ii) the boys are always separated?
- (iii) the girls and boys occupy alternate places?
- (iv) the boys are always separated by exactly 2 girls?

Solution

(i) Grouping Method

Since the 2 boys are always together, we consider them as 1 unit.
Therefore, any arrangement would be like the following.

$B_1 B_2$ G_1 G_2 G_3

Take note of this
"grouping together
as 1 unit" method.

Number of ways to arrange this unit and the 3 girls in a row = $4!$.

Number of ways to arrange the 2 boys within the unit = $2!$.

Required number of ways = $4! \times 2! = 48$.

(ii) Method 1: Complement Method

Number of ways to arrange 3 girls and 2 boys with no restriction = $5!$

Number of ways in which the boys are always separated

= Number of ways to arrange 3 girls and 2 boys with no restriction

– Number of ways in which the 2 boys are always together

= $5! - 48 = 72$.

Answer for this
was found in (i)

Method 2: Slotting Method

↑ G_1 ↑ G_2 ↑ G_3 ↑

Number of ways to arrange the 3 girls = $3!$.

For the 2 boys to be separated, they can occupy any 2 of the 4 slots shown by ↑

Number of ways to arrange the 2 boys = ${}^4C_2 \times 2!$ or ${}^4P_2 = 12$.

4C_2 ways to select 2 slots from 4.
 $2!$ ways to arrange the 2 boys in
the 2 chosen slots.

Required number of ways = $3! \times 12 = 72$.

Remarks:

When there are more than 2 objects that need to be separated, use Method 2 instead of Method 1 as it would be more straightforward. Method 1 can be tricky.

(iii) Arrangement must be GBGBG.

Number of ways to arrange the 3 girls = $3!$.

Number of ways to arrange the 2 boys = $2!$.

Required number of ways = $3! \times 2! = 12$.

(iv) Case 1: BGGBG

Number of ways = $3! \times 2! = 12$. (Same argument as (iii))

Case 2: GBGGB

A similar argument gives 12 ways.

Total number of ways = $12 + 12 = 24$.

Note that the 2 cases are
mutually exclusive, thus the
use of Addition Principle.

Example 11

Find the number of ways in which a committee of 4 can be formed from 6 boys and 5 girls if

- (i) it must contain 2 boys and 2 girls;
- (ii) it must contain at least 1 boy and 1 girl;
- (iii) either the oldest boy or the oldest girl must be included, but not both.

Solution

(i) Number of ways of choosing the 2 boys = 6C_2 .

Number of ways of choosing the 2 girls = 5C_2 .

Required number of ways = ${}^6C_2 \times {}^5C_2 = 150$

(ii) **Method 1: Systematic Listing.**

Number of boys	Number of girls	Number of ways
1	3	${}^6C_1 \times {}^5C_3 = 60$
2	2	${}^6C_2 \times {}^5C_2 = 150$ (from part (i))
3	1	${}^6C_3 \times {}^5C_1 = 100$

Total number of ways = $60 + 150 + 100 = 310$

Remarks:

${}^6C_1 \times {}^5C_1 \times {}^9C_2$ is incorrect as there is double-counting.

For eg, $B_1 G_1 B_2$ and

$B_2 G_1 B_1$ are the same grouping but are considered as two distinct cases in the computation of ${}^6C_1 \times {}^5C_1 \times {}^9C_2$.

Method 2: Complement Method.

Finding the number of committees containing at least 1 boy and 1 girl is the same as finding the number of committees without restriction and then subtracting the number of all-boy and all-girl committees.

Number of committees without restriction = ${}^{11}C_4$

Number of all-boy committees = 6C_4 .

Number of all-girl committees = 5C_4 .

Number of committees with at least 1 boy and 1 girl = ${}^{11}C_4 - {}^6C_4 - {}^5C_4 = 310$.

(iii) Case 1: Oldest boy included, oldest girl excluded

Since the oldest boy is included, we only need to choose another 3 members from the remaining members.

Number of remaining members to choose from = $6 + 5 - 1 - 1 = 9$.

Number of ways = ${}^9C_3 = 84$.

Case 2: Oldest girl included, oldest boy excluded

A similar argument gives 84 ways.

Total number of ways = $84 + 84 = 168$.

Note that the 2 cases are mutually exclusive.

Example 12

How many ways can 4 Physics books and 3 Mathematics books be arranged on a shelf if a selection is made from 6 different Physics books and 5 different Mathematics books?

Solution

Consider the selection of the 7 books first.

Number of ways to choose 4 Physics books = 6C_4 .

Number of ways to choose 3 Mathematics books = 5C_3 .

Number of ways of choosing the books = ${}^6C_4 \times {}^5C_3 = 150$.

Now, number of ways to arrange each selection of 7 books = $7!$.

Total number of possible arrangements = $150 \times 7! = 756000$.

Why isn't the answer ${}^6P_4 \times {}^5P_3$

Example 13

How many different selections can be made by taking some or all of the letters

(i) a, b, c, d, e ?

(ii) a, a, a, b, b, c ?

Solution**(i) Method 1**

We can select either only 1, or 2 or 3 or 4 or all 5 letters.

$$\text{Number of non-empty selections} = {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 31.$$

Method 2

Consider the letter 'a'. It can either be chosen or not chosen

Number of ways to choose 'a' = 2.

A similar argument holds for the other 4 letters.

$$\text{Number of selections} = 2 \times 2 \times 2 \times 2 \times 2 = 2^5.$$

However in one of these selections, all the letters were not chosen.

$$\therefore \text{Number of non-empty selections} = 2^5 - 1 = 32 - 1 = 31.$$

Notice that method 1 and 2 give the same answer, which means that ${}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 2^5 - 1$. The interesting link between them can be explained by the Binomial Theorem of expansion $(1+x)^5$ with $x=1$.

Try it yourself!

(ii) A selection can either contain 3 'a's, 2 'a's, 1 'a' or no 'a'.

Number of ways to choose 'a' = 4.

Similarly, number of ways to choose 'b' = 3.

and number of ways to choose 'c' = 2.

$$\text{Number of non-empty selections} = 4 \times 3 \times 2 - 1 = 24 - 1 = 23.$$

Example 14 HCI J2CommonTest1 9740/2009/Q2

There are three schools and six newly graduated teachers, including one married couple.

Each teacher is to be posted to one of the schools.

Find the number of ways the postings can be done

- (i) if there are no restrictions,
- (ii) if every school must receive exactly two teachers and the couple cannot be posted to the same school,
- (iii) if every school must receive at least one teacher.

Solution

- (i) For each teacher, there are 3 possible choices (i.e. 3 schools)

$\therefore$ Number of ways if there are no restrictions = $3^6 = 729$.

- (ii) **Method 1 : Complement method**

Number of ways for each school to receive exactly 2 teachers = ${}^6C_2 \times {}^4C_2 = 90$.

Number of ways each school receives exactly 2 teachers and the couple is posted to the same school = $3 \times {}^4C_2 = 18$.

Using the Complement Method, required number of ways = $90 - 18 = 72$.

Method 2

There are 4C_1 ways to pair a teacher with the husband and 3C_1 ways to pair a teacher with the wife. The remaining 2 teachers will be paired together.

The 3 pairs can be posted to the 3 schools in $3!$ ways.

$\therefore$ Number of ways if every school receives exactly two teachers and the couple is not posted to the same school = ${}^4C_1 \times {}^3C_1 \times 3! = 72$.

- (iii) **Case 1:** Two schools each have 1 teacher, one school has 4 teachers

Number of ways = $\frac{{}^6C_1 \times {}^5C_1 \times {}^4C_4}{2!} \times 3! = 90$.

There are $3!$ ways to assign the 3 groups of teachers to 3 schools

Case 2: One school has 1 teacher, one school has 2 teachers, one school has 3 teachers.

Number of ways = ${}^6C_1 \times {}^5C_2 \times {}^3C_3 \times 3! = 360$.

Case 3: Each of the three schools has 2 teachers

Number of ways = $\frac{{}^6C_2 \times {}^4C_2 \times {}^2C_2}{3!} \times 3! = 90$.

$\therefore$ Required number of ways = $90 + 360 + 90 = 540$.

Example 15 SRJCPrelim/9740/2013/02/Q6

Mr Ahmad, Miss Banu and the families of Lim and Tan were invited to a wedding banquet. The Lim family consists of a man and both his parents while the Tan family consists of a woman and both her parents. The invited guests sit at a round table for eight. Find the number of ways of arranging the eight people if

- (i) there are no restrictions;
- (ii) the men and women are seated alternately;
- (iii) Mr Ahmad and Miss Banu sit separately, and the members of the same family are seated together;
- (iv) the members of the same family are seated together and the seats are numbered.

Solution

(i) Number of arrangements = $(8 - 1)! = 5040$

(ii) Number of arrangements = $(4 - 1)! \times 4! = 144$

Explanation:

No of ways to arrange the women around the table first = $(4 - 1)!$

To separate the men, “slot in” the men, no of ways = ${}^4C_4 \times 4!$

$\therefore$ Number of arrangements = $(4 - 1)! \times 4! = 144$

Note: Can also consider arrange the men first and “slot in” the women.

(iii) Slotting Method

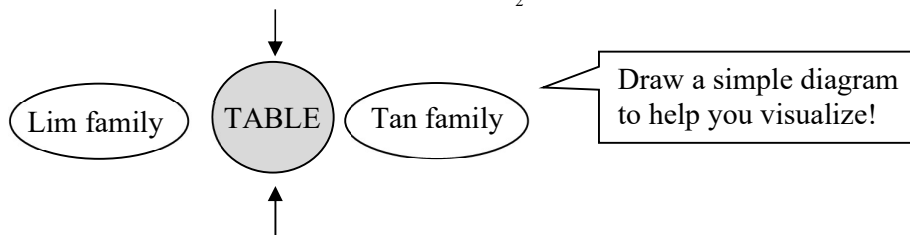
Number of arrangements = $[(2 - 1)! \times 3! \times 3!] \times 2! = 72$

Explanation:

For the members of the same family to be seated together, we group Tan family and Lim family as one unit each, each consisting of three members. Since we want to separate Mr Ahmad and Ms Banu, we leave them out first and “slot in” later.

No of ways to arrange Tan family and Lim family around the table
 $= (2 - 1)! \times 3! \times 3!$

No of ways to “slot in” Mr Ahmad and Ms Banu = ${}^2C_2 \times 2! = 2!$



$\therefore$ Number of arrangements = $[(2 - 1)! \times 3! \times 3!] \times 2! = 72$

Complement Method

$$\begin{aligned}\text{Number of arrangements} &= [(4-1)! \times 3! \times 3!] - [(3-1)! \times 3! \times 3! \times 2!] \\ &= 216 - 144 = 72\end{aligned}$$

Explanation:

The complement case is that Mr Ahmad and Ms Banu are seated next to each other.

Using Grouping Method, we consider Mr Ahmad and Ms Banu as one unit.

Also, Tan family and Lim family are two units that consists of three members each.

$$\text{No of ways to arrange the 3 units} = (3-1)! \times 3! \times 3! \times 2!$$

Total no of ways without restriction on Mr Ahmad and Ms Banu, while keeping Tan and Lim with their respective family = $(4-1)! \times 3! \times 3!$ since there are 4 units: Mr Ahmad, Ms Banu, Tan family and Lim family.

$$\therefore \text{Number of arrangements} = [(4-1)! \times 3! \times 3!] - [(3-1)! \times 3! \times 3! \times 2!] = 72$$

(iv) $\text{Number of arrangements} = (4-1)! \times 3! \times 3! \times 8 = 1728$

Explanation:

When the members of the same family are seated together, we have 4 units in total: Mr Ahmad, Ms Banu, Tan family and Lim family.

$$\text{No of ways to arrange the 4 units at round table} = (4-1)! \times 3! \times 3!$$

Since the seats are now numbered, we multiply by number of seats, which is 8.

$$\therefore \text{Number of arrangements} = (4-1)! \times 3! \times 3! \times 8 = 1728$$

Example 16 ACJCPrelim9233/2006/01/Q12

Belle and Candy went out with eight other friends to have lunch at a restaurant. They were shown to two round tables: one a four-seater, and the other a six-seater.

- (i) Find the number of ways in which all ten can be seated.
- (ii) If Belle and Candy wish to sit next to each other, find the number of arrangements which would make this possible.

Solution

- (i) Firstly, choose 4 out of the 10 people to occupy the 4-seater, the remaining 6 people occupy the 6-seater. No of ways = $^{10}C_4$.

No of ways to arrange the 4 people around 4-seater = $(4-1)!$

Similarly, no of ways to arrange the 6 people around 6-seater = $(6-1)!$

$\therefore$ Number of ways = $^{10}C_4 \times (4-1)! \times (6-1)! = 151200$.

- (ii) Case 1: Belle and Candy sit at the 4-seater.

Number of ways = $\left[^8C_2 \times (3-1)! \times 2! \right] \times (6-1)! = 13440$.

Case 2: Belle and Candy sit at the 6-seater.

Number of ways = $\left[^8C_4 \times (5-1)! \times 2! \right] \times (4-1)! = 20160$.

Total number of arrangements = $13340 + 20160 = 33600$.

Explanation:

Since Belle and Candy sit next to each other, consider them as one unit. They are either seated at the 4-seater or the 6-seater.

Case 1:

If they sit at the 4-seater, we need to choose 2 more people to sit with them at the 4-seater.

No of ways to choose these people = 8C_2 .

Since Belle and Candy now form one unit, there are a total of 3 units to be arranged at the 4-seater, no of ways = $(3-1)! \times 2!$.

The no of ways to arrange the remaining 6 people around the 6-seater remains as $(6-1)!$.

$\therefore$ Required number of ways = $\left[^8C_2 \times (3-1)! \times 2! \right] \times (6-1)! = 13440$.

No of ways for Case 2 can be found using the same approach as Case 1.

CONCLUSION

The first known publication on the study of permutations and combinations was a book titled “*Ars Conjectandi*”, attributed to James Bernoulli posthumously in 1713 and inspired by an exchange of letters between Pierre de Fermat and Blaise Pascal on the topic. At once a training in logical and systematic thinking as well as a stepping stone towards the study of combinatorics and probability and statistics, this exercise in counting also has influences on everyday decisions such as whether one should try to remember the orders for drinks for a group of 8 people.

APPENDIX A

Result 4

The number of permutations of n objects, taken all at a time without replacement, of which n_1 are of the 1st type, n_2 of the 2nd type, $\dots$, n_k of the k^{th} type, where

$n_1 + n_2 + \dots + n_k = n$ is given by $\frac{n!}{n_1!n_2!\dots n_k!}$.

Question: Why is there a need to divide by $n_1!n_2!\dots n_k!$?

Let's consider a simple case:

How many ways can we arrange the 3 letters AAB?

If all 3 letters were distinct, then the number of permutations would simply be $3!$.

But this involves some double-counting here.

In the $3!$ permutations, we actually have $2!$ permutations of the form AAB.

If we write AAB as A_1A_2B , we can see that the $2!$ identical permutations arise from the following arrangements

$$A_1A_2B \text{ and } A_2A_1B$$

(Note that $2!$ is the number of ways of arranging A_1A_2)

Similarly, we have $2!$ permutations of the form ABA and another $2!$ permutations of the form BAA. Because we have 2 identical letters, every permutation will occur $2!$ times within the total number of $3!$ permutations.

Therefore, the number of distinct permutations of the 3 letters AAB is $\frac{3!}{2!} = 3$.

APPENDIX B**Result 5**

The number of permutations of n DISTINCT objects in a circle is given by $\frac{n!}{n} = (n-1)!$.

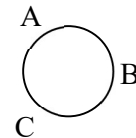
Question: Why is there a need to divide by n ?

Let's consider the following question first:

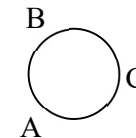
Is there a relationship between permutations in a line and in a circle?

We can form a circular permutation from a linear permutation by joining its last object to the first. Consider the following permutations of the 3 letters ABC. If we join the last letter to the first in each permutation by bending the line around to form a circle, we would get the following circular permutations.

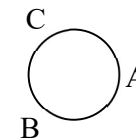
A ——— B ——— C



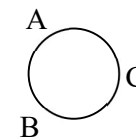
B ——— C ——— A



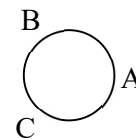
C ——— A ——— B



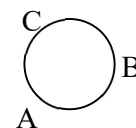
A ——— C ——— B



B ——— A ——— C



C ——— B ——— A



Notice that the first 3 circular permutations are identical to one another, and the last 3 circular permutations are identical to one another, but they are not identical across the 2 groups.

Therefore there are only $\frac{3!}{3} = 2$ distinct circular permutations of the 3 letters ABC.

SUMMARY